

College Management System

Product Requirements Document (PRD)

1. Project Overview

Project Name

College Management System

Purpose

The College Management System is a web-based platform designed to automate academic and administrative operations within a college. The system reduces manual work, minimizes human errors, and improves efficiency in managing students, staff, subjects, marks, notices, and leave requests.

2. User Roles

Admin

Full control over the entire system.

Staff

Academic management role with restricted permissions.

Student

View-oriented role with limited interaction capabilities.

3. Technology Stack

Frontend

- HTML
- CSS

Backend

- Django

Database

- SQLite

JavaScript

- Not required
- Application should function primarily through Django server-side rendering

Responsive Design

- Desktop
 - Tablet
 - Mobile
-

4. Authentication System

Admin Authentication

Login

- Username
- Password

Permissions

- Create Admin accounts
- Manage all system data
- Reset Student credentials
- Reset Staff credentials

Multiple Admin Support

- Supported
-

Staff Authentication

Account Creation

- Created only by Admin

First Login

- Staff ID
- Date of Birth

First Login Requirement

- Must create password

Future Logins

- Staff ID
- Password

Restrictions

- Cannot change password after setup
 - Cannot edit profile
-

Student Authentication

Account Creation

- Created only by Admin

Login

- Roll Number
- Date of Birth

Restrictions

- Cannot change password
 - Cannot edit profile
-

5. Student Management Module

Student Data Fields

- Roll Number (Auto Generated)
- Full Name
- Date of Birth
- Gender
- Email
- Phone Number
- Address
- Parent/Guardian Name
- Parent Phone Number
- Course
- Department
- Semester

- Year of Admission
- Profile Photo

Permissions

Admin

- Create Student
- View Student
- Update Student
- Delete Student

Staff

- View Student Records Only

Student

- View Own Profile Only
-

6. Staff Management Module

Staff Fields

- Staff ID
- Full Name
- Email
- Phone Number
- Department
- Designation
- Date of Birth

Permissions

Admin

- Create Staff
- Edit Staff
- Delete Staff
- Assign Subjects

Staff

- View Own Profile
-

7. Department Management Module

Permissions

Admin

- Create Department
- View Department
- Update Department
- Delete Department

Staff

- View Only

Students

- View Own Department
-

8. Course Management Module

Permissions

Admin

- Create Course
- View Course
- Update Course
- Delete Course

Staff

- View Only

Students

- View Own Course
-

9. Subject Management Module

Features

Admin

- Create Subject

- Edit Subject
- Delete Subject
- Assign Subject to Staff

Rules

- One Staff can teach multiple Subjects
 - One Subject can be assigned to only one Staff
-

10. Semester Management

Student Assignment

Every Student must belong to:

- Course
- Department
- Semester

Semester Promotion

Admin Dashboard shall contain:

Promote All Students Button

Function:

- Semester 1 → Semester 2
- Semester 2 → Semester 3
- Semester 3 → Semester 4
- Semester 4 → Semester 5
- Semester 5 → Semester 6

Restrictions:

- Semester 6 students are not promoted further
-

11. Marks Management Module

Structure

Each Student has:

- One Total Mark per Subject

Staff Permissions

Staff can:

- Add Marks
- Edit Marks
- Delete Marks

Restriction:

- Only for Subjects assigned to them

Admin Permissions

Admin can:

- View All Marks
- Add Marks
- Edit Marks
- Delete Marks

Student Permissions

Students can:

- View Marks
- Download Mark Sheet

12. Notice Management Module

Notice Creators

- Admin
- Staff

Notice Viewers

- Students Only

Notice Targets

- Entire College
- Specific Course
- Specific Department
- Specific Semester

Notice Fields

- Title
- Description
- Target Audience
- Created By
- Creation Date
- Expiry Date

Attachments

Not Supported

Expiry Logic

Expired notices are automatically hidden.

13. Leave Request Module

Leave Request Fields

- Student Name
- Roll Number
- From Date
- To Date
- Reason
- Attachment

Workflow

1. Student creates leave request.
2. Student selects Staff recipient.
3. Request is sent to selected Staff.
4. Staff reviews request.
5. Staff approves or rejects request.
6. Student receives updated status.

Staff Permissions

- View Requests
- Approve Requests
- Reject Requests
- Add Remarks

Student Permissions

- Submit Request

- View Status
- Cancel Pending Request

Admin Permissions

- View All Leave Requests
-

14. Dashboard Requirements

Student Dashboard

Display:

- Profile Information
 - Marks
 - Notices
 - Leave Requests
-

Staff Dashboard

Display:

- Assigned Subjects
 - Student List
 - Marks Management
 - Notices Management
 - Leave Requests
 - Profile Information
-

Admin Dashboard

Display access to all modules:

- Students
- Staff
- Admins
- Departments
- Courses
- Subjects
- Marks
- Notices
- Leave Requests

Admin has complete CRUD access throughout the system.

15. Non-Functional Requirements

Security

- Role-Based Access Control
- Password Hashing using Django Authentication
- CSRF Protection
- Session Management
- Secure File Upload Validation

Performance

- Fast page loads
- Optimized database queries

Usability

- Mobile Responsive
- Simple Navigation
- Clean Interface

Reliability

- Data consistency
 - Prevent unauthorized access
-

16. Database Entities

Core Tables

- Admin
 - Staff
 - Student
 - Department
 - Course
 - Subject
 - SubjectAssignment
 - Marks
 - Notice
 - LeaveRequest
-

17. Future Enhancements

- Attendance Management

- Fee Management
 - Timetable Management
 - Report Generation
 - Library Module
 - Placement Module
 - Hostel Module
 - Transport Module
-

18. Success Criteria

- Complete student lifecycle management
- Accurate marks management
- Efficient notice distribution
- Streamlined leave request workflow
- Reduced manual administrative workload
- Fully responsive user experience